Transparent ESG Prediction

ENVIRONMENTAL

SOCIAL

GOVERNANCE

Machine Learning for Accessible Corporate Sustainability Assessment

Caden – John – Ethan | Syracuse University | IST 707

Problem & Stakeholder

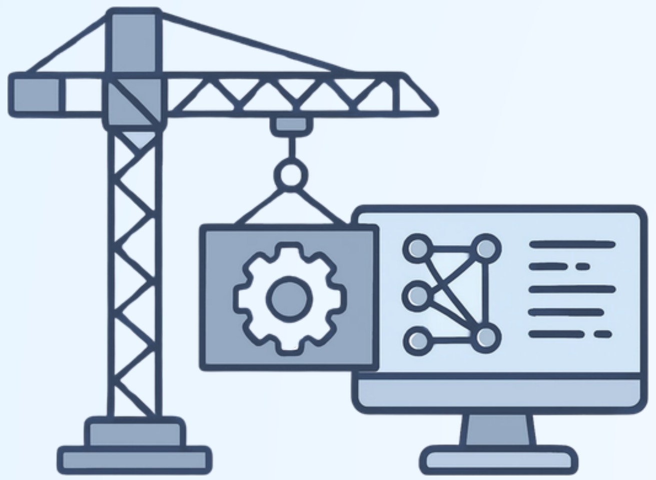
The Problem

- Commercial ESG ratings cost $30,000+/year
- Locks retail investors out of a $30T market
- Most commercial ESG ratings are a "black box" with no public methodology
- Even transparent providers rely on 100+ proprietary data sources unavailable for free

Primary Stakeholder: Retail Investors

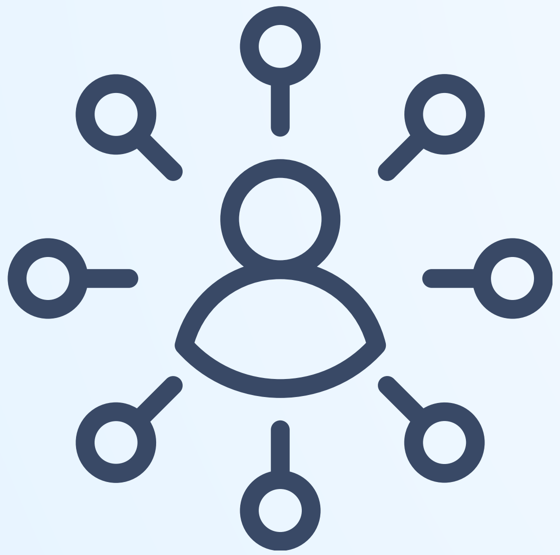
- Want values-aligned portfolios but can't afford institutional subscriptions
- No free, transparent alternative exists

Envisioned Solution



What We're Building

- Free ESG prediction system using only public SEC 10-K filings and financial data
- FinBERT extracts sentiment and language patterns from ESG-relevant text
- ElasticNet model predicts Environmental, Social, and Governance scores



How It Connects to Stakeholders

- Retail investors get free, accessible ESG assessments
- SHAP explainability and ElasticNet coefficients will show why a company scores the way it does, addressing the industry-wide "black box" problem

Data

323

Companies

MSFT, AAPL, AMZN, etc.

4

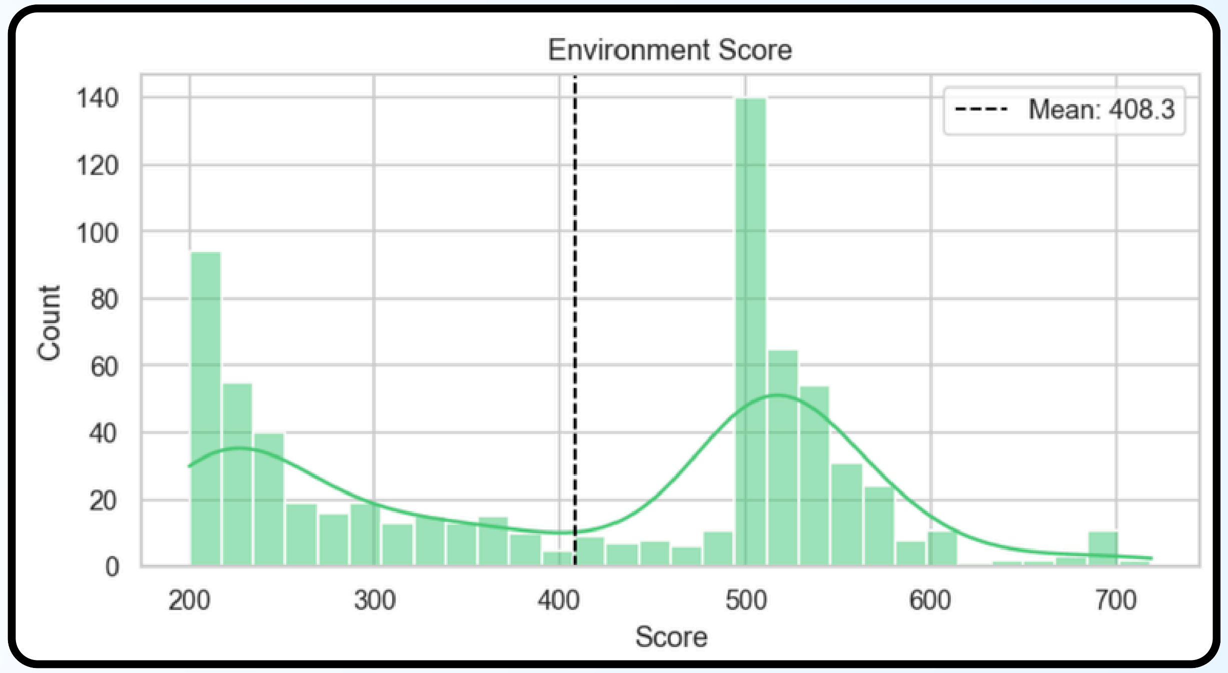
Targets

Environmental, Social, Governance, and Total

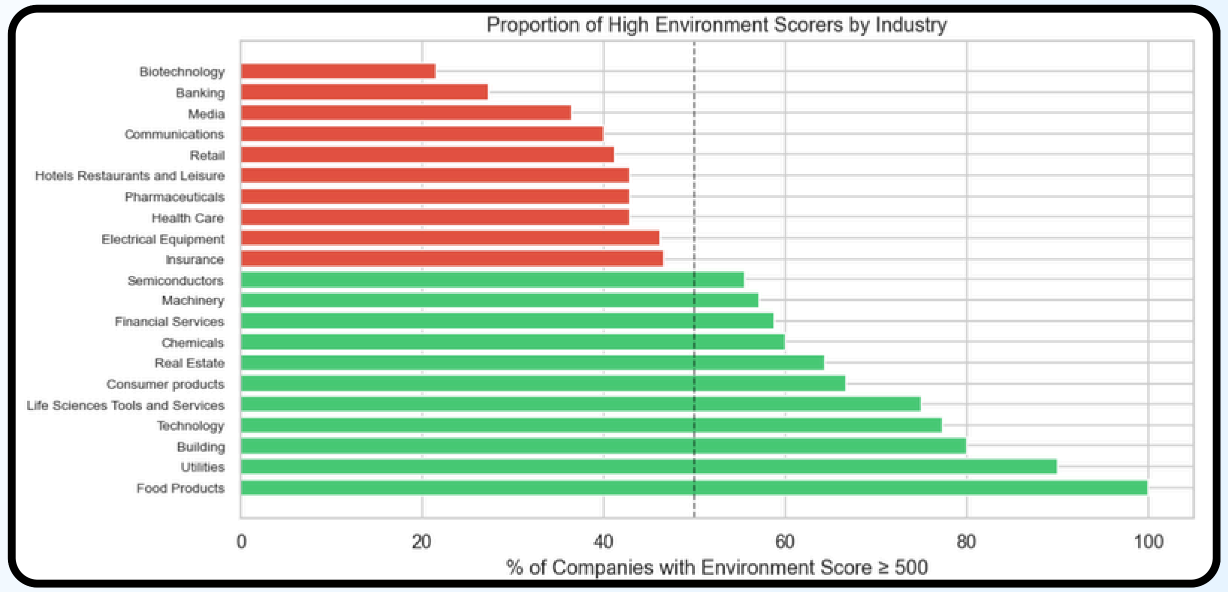
108

Features

FinBERT, Keywords, Structure, Financials



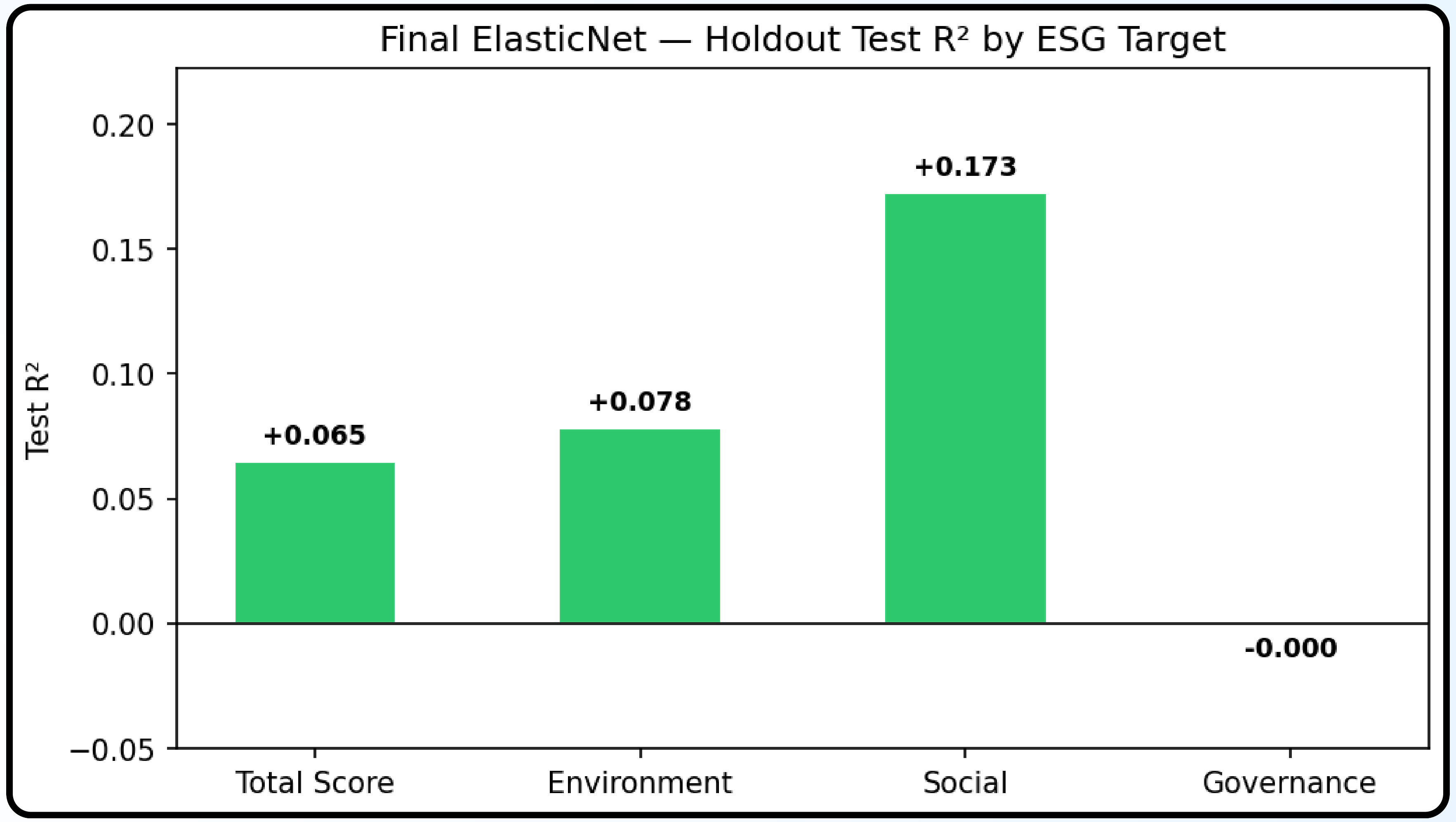
- **ESG Ratings:** Numerical scores and letter grades across 3 pillars plus a total score (FY2021)
- **10-K Filings:** Business, Risk Factors, and MD&A sections extracted per filing
- **FinBERT Features:** Sentiment scores from ESG-relevant sentences via a finance-specific language model
- **SEC Financials:** Total assets, public float, debt-to-assets, profit margin, return on equity



Initial Results

ElasticNet Model Evaluation

- **Sparse Signal:** ElasticNet selected only 27 to 31 of 108 features, confirming sparse signal structure
- **Best Performer:** Social explains 17.3% of score variance with 31 features
- **Structural Ceiling:** Governance had 0 features selected, confirming board and shareholder data are not captured in 10-K filings

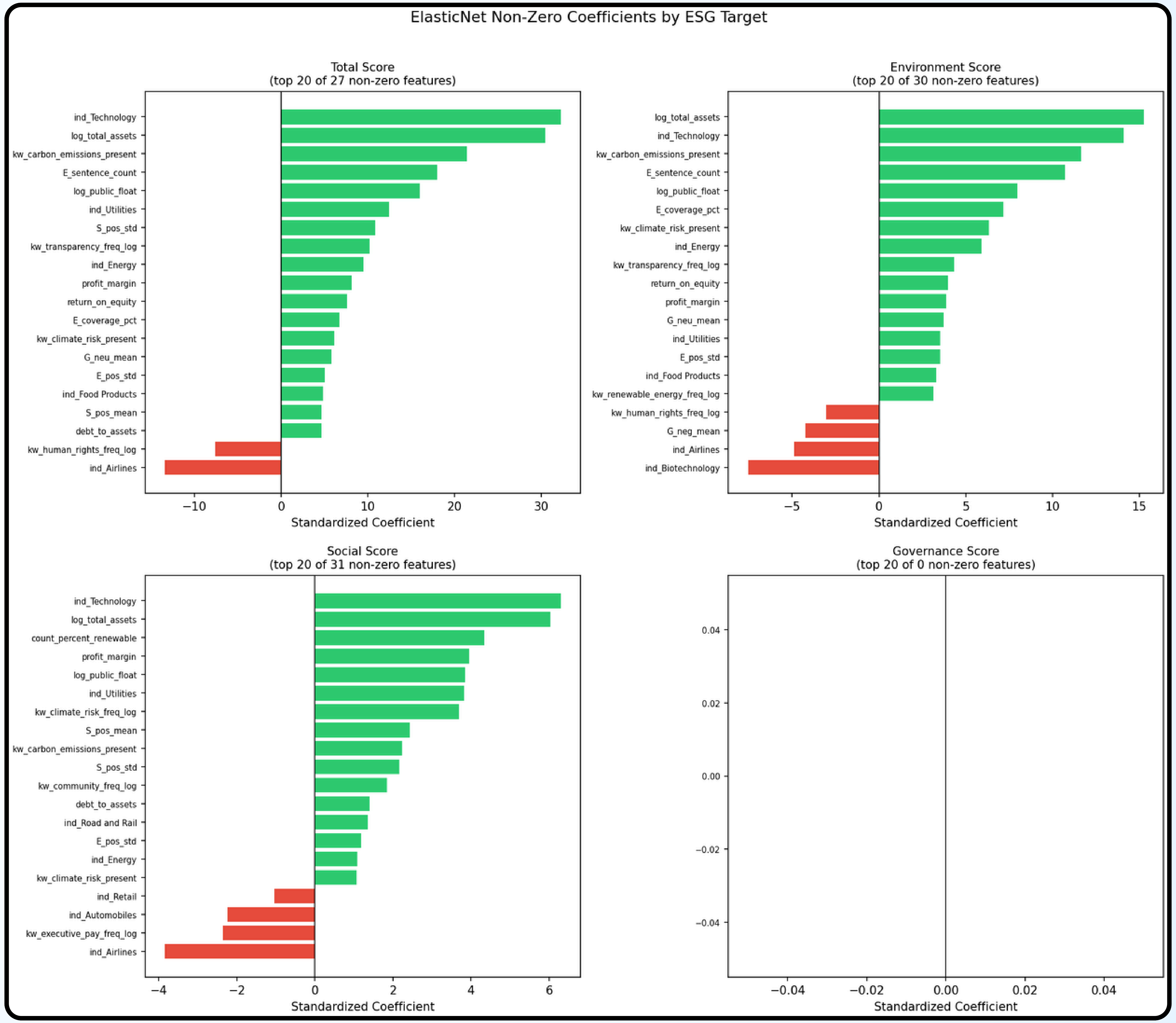


TARGET	TEST R^2	MAE	SCORE STD	FEATURES USED
Social best	0.173	31.45	55	31 / 108
Environment	0.078	104.10	143	30 / 108
Total	0.065	145.66	216	27 / 108
Governance ceiling	~0.000	36.33	48	0 / 108

Initial Results (cont.)

ElasticNet Model Findings

- **Dominant Drivers:** Company size and industry dummies are the strongest predictors across all predictable pillars
- **Text Features:** FinBERT sentiment and keyword features contribute meaningful coefficients across Social and Environment pillars
- **Governance Finding:** Empty coefficient column confirms a structural ceiling, not a modeling failure



Challenges

Addressed

- **XBRL Contamination:** 9 companies removed where raw financial code was extracted instead of text
- **Bimodal Distribution:** Score split (most pronounced in environment) identified as a disclosure artifact, not a genuine performance signal
- **Multicollinearity:** Structured features reduced from 9 to 5 to limit redundancy

Currently Facing

- **Sentence Cap:** FinBERT capped at 100 sentences; 68-79% of companies hit the limit, cutting off Social/Governance signal
- **Governance Ceiling:** Board composition and shareholder data needed for governance are absent from 10-K filings
- **Data Access:** ESG Enterprise draws on paid news, NGO, and controversy sources our free pipeline cannot replicate

Plans & Goals

YOU ARE HERE

Feature Improvements

- **Raise FinBERT sentence cap 100 → 300+** *(68–79% of companies hit cap for Social/Governance)*
- **Add section-presence binary features** *(structural disclosure signal)*

Final Model Run

- Re-run final validation with **improved feature set**
- Add **5-fold CV evaluation** per pillar to validate holdout results
- Improve all four ESG targets (***best: Social $R^2 = 0.173$***)

Explainability

- Document **ElasticNet coefficients** and **SHAP findings** across all four pillars
- Finalize **governance structural ceiling** conclusion
- **Refine grade classifier** and improve the letter grade accuracy

Final Deliverable

- **Write final report** covering methodology, results, and limitations
- **Proof of concept:** publicly available data can partially replicate ESG ratings transparently